



Literature Results

Objective

I'm a solo student engineer building a submission for the Razorpay AI Buildathon 2026 (deadline September 5, 2026). I have about one week alone, and need to build using existing pretrained models, open datasets, and public APIs only — no training a model from scratch.

Search academic papers, preprints, patents, and industry/security research reports (not just clinical/biomedical sources) from the last 18 months for documented, still-unresolved problems at the intersection of payments, fraud, checkout, refunds/chargebacks, or AI agents transacting on behalf of humans. I need candidates that are stated as unsolved gaps in the sources themselves, not problems I'm assuming exist.

The buildathon has five tracks (<https://razorpay.com/buildathon/>): AI Growth & Agentic Commerce (grow merchant revenue, make merchants sellable to AI buyers), AI Risk Manager (stop the merchant losing money to fraud, returns, chargebacks — detection/defense only, not attack-building), AI Revenue Recovery (payment failures, checkout abandonment, overdue receivables), AI Finance Controller (reconciliation, exception handling), and an Open Track.

For each candidate you find, report:

- The specific source(s) stating the problem is unresolved, with citation and date
- Which track it fits and why, quoting the track's own language
- What dataset, benchmark, or API exists to build and evaluate a solution — name it specifically with a link, don't say data "likely exists"
- What a demo of it would visually show a judge in 30 seconds

Exclude these spaces — I've already confirmed they're crowded or already owned by an incumbent: hospital-bill auditing against CGHS/PM-JAY rates (BillOkay, AuditorX, eVaakil already do this); generic RTO/chargeback-risk prediction (Razorpay's own Thirdwatch and Chargeback Shield already do this); mule-merchant/transaction-laundering detection (PayLegit, BioCatch already do this); AI shopping agent identity or spend-mandate verification (already being standardized by Visa Trusted Agent Protocol, Mastercard Agent Pay, Google's AP2, OpenAI/Stripe's ACP).

Return 5-8 ranked candidates, most promising first, with sources for every claim.

Results

Ranked build candidates

Scope and screening

I treated “last 18 months” relative to the stated deadline as **5 March 2025–5 September 2026**. I excluded the four spaces named in the prompt. I also excluded proposals whose core contribution would be agent identity or spend-mandate verification; merchant-side transaction-state validation, recovery, and defensive monitoring remain in scope.

The ranking weights: **explicitness of the unresolved gap, one-week feasibility, differentiation from incumbents, availability of public evaluation resources, and 30-second demo clarity**. Patent evidence is used conservatively: only patents that explicitly criticize existing approaches support an unresolved-gap claim; a patent’s existence alone is not treated as proof.

Rank	Candidate name	Primary track	Source-stated unresolved gap	Named public build/evaluation resource	30-second demo
1	Stateful refund/checkout workflow guard	AI Growth & Agentic Commerce	Agent systems still lack "reversible commitment lifecycles," "cancellation and refund callbacks," split-state detection, and recovery-aware evaluation; task success misses these failures ([1.1], [1.2], [1.3], [2.1])	ThinkingBox / ThinkingBox-Bench — https://github.com/microsoft/thinkingbox ; Agentic Commerce Blueprint v0.9.2 — https://github.com/dmsfiris/agentic-commerce-blueprint/releases/tag/v0.9.2	Agent books/updates order, guard catches unsafe refund or stale checkout step, asks for revalidation, then safely completes
2	Merchant-side stale-state and prompt-injection firewall	AI Risk Manager	Merchant-side decision freshness is unresolved: protocols alone cannot stop stale price/inventory/policy use; "cryptographic mandates alone cannot protect against prompt injection or malicious product content"; production security/live interoperability not yet tested ([3.1], [3.2], [4.1])	Agentic Commerce Blueprint — https://github.com/dmsfiris/agentic-commerce-blueprint ; OWASP GenAI Security Project — https://genai.owasp.org/	Product page is tampered; firewall blocks provider call before payment, shows stale hash / hostile content / re-check result
3	Evidence-grounded multi-document reconciliation copilot	AI Finance Controller	LLMs remain weak on reconciliation: best exact final balance-sheet accuracy only 46%; document-linking failures dominate; synthetic benchmark still misses real OCR/document messiness ([5.1], [5.2], [5.3])	FinBalance — https://doi.org/10.48550/arxiv.2606.15949 ; FinMaster — https://doi.org/10.48550/arxiv.2505.13533	Upload invoice, memo, bank line, ledger; copilot proposes entry, cites evidence, and highlights wrong linkage/conflict
4	Many-to-many cash-application exception resolver	AI Finance Controller	Open reproducible datasets are missing; many-to-many matching remains hard when one payment clears many invoices; recall degrades 2–5 pp after bank-format drift; batched remittance/partial settlement exceptions remain unresolved ([6.1], [6.2], [6.3], [7.1])	Razorpay RouteX / Payments / Webhooks sandbox APIs — https://razorpay.com/docs/api/ ; FinBalance for reconciliation-style eval — https://doi.org/10.48550/arxiv.2606.15949	One remittance hits many invoices; resolver clusters candidates, assigns confidence, and routes only ambiguous tails to human queue
5	Agent-readable catalog/checkout readiness linter	AI Growth & Agentic Commerce	Merchants need new agent-facing infrastructure: products may be discoverable but "not checkout-ready"; small-publisher/merchant adoption barriers, multi-agent attribution/billing, and live interoperability remain open ([3.1], [8.1])	Amazon Shopping Queries Dataset / ESCI — https://github.com/amazon-science/esci-data ; Schema.org Product spec — https://schema.org/Product ; Google Rich Results Test — https://search.google.com/test/rich-results	Paste a PDP URL; linter scores discoverability, machine-readable attributes, checkout affordances, and shows fixes turning red to green
6	Pre-chargeback evidence and merchant-cardholder clarification workspace	AI Revenue Recovery	Current dispute handling is inefficient; pre-chargeback communication is meant to resolve issues before processor escalation, reducing chargebacks and network load; refund/dispute auditability remains a gap in agentic commerce ([9.1], [10.1])	CFPB Consumer Complaint Database / API — https://www.consumerfinance.gov/data-research/consumer-complaints/ ; Patent-inspired workflow for pre-chargeback messaging eval ([9.1])	Complaint arrives; workspace drafts merchant clarification, requests missing proof, assembles receipt timeline, and shows chargeback avoided

Rank	Candidate name	Primary track	Source-stated unresolved gap	Named public build/evaluation resource	30-second demo
7	POS terminal attestation anomaly console	AI Risk Manager	Existing terminal defenses are bypassable: passive/active attacks, weak app isolation, and limited visibility into concurrent fraud remain; complete tamper-proofing is impossible, so dynamic attestation/feedback loops are needed ([11.1], [11.2], [11.3], [11.4])	IoT-23 dataset — https://www.stratosphereips.org/datasets-iot23 ; TON_IoT — https://research.unsw.edu.au/projects/toniot-datasets ; patent-derived attestation event model ([11.1], [11.4])	Console streams terminal attestations; one device turns anomalous, risk spikes, terminal is quarantined before next payment

Table: This table summarizes seven shortlisted Buildathon ideas ranked by promise, with exact-track mapping, concise source-stated unresolved gaps, concrete public build resources, and a judge-friendly 30-second demo concept.

1. Stateful refund and checkout workflow guard

Problem stated by sources. *From Super-Apps to Agent Economies* (April 2026) says action completion is advancing before payment, liability, and dispute frameworks, and specifically identifies missing reversible commitment lifecycles, cancellation/refund callbacks, dispute routing, linked receipts, split-state detection, and recovery mechanisms. It further says current evaluation overweights task completion while neglecting authorization bounds, reversal, receipt preservation, and human recovery burden. ([1.4], [1.1], [1.2])

The measured problem is substantial. *Beyond Task Success* (July 2026) found that **10 of 18 models skipped a mandatory payment-confirmation checkpoint**, despite near-perfect ordinary task-success scores. *ThinkingBox* (August 2026) evaluates refunds and other stateful workflows: its strongest model reached **65.36% pass@1 but only 25.25% all-20 reliability**, and failed runs could terminate cleanly after apparently valid state-changing calls. ([2.1], [12.1])

Track fit. Best fit: **AI Growth & Agentic Commerce**, specifically “**make merchants sellable to AI buyers.**” An AI buyer cannot safely transact with a merchant unless checkout, cancellation, and refund state transitions are machine-operable and recoverable. Secondary fit is Open Track.

Build/evaluation resources. Use Microsoft’s open [ThinkingBox/ThinkingBox-Bench](#), which contains 507 policy-conditioned workflows including retail, bookings, and refunds, and the open [Agentic Commerce Blueprint v0.9.2](#). Connect a tiny mock shop to [Razorpay’s test-mode Payments and Refund APIs](#). The source reports 66/66 tests for the Blueprint but explicitly does not establish production security or population error rates. ([3.1], [3.3], [12.1])

One-week MVP. Do not build a shopping agent. Build middleware around an existing model: a transaction state machine, invariant checker, event/receipt graph, and human-recovery panel. Evaluate baseline versus guarded agent on 20–40 adapted ThinkingBox refund/checkout cases.

30-second demo. Left pane: an agent changes an order and requests a refund. A payment succeeds while order state remains stale. The guard turns the flow red, identifies the split state, blocks the next mutation, reconstructs the receipt trail, and offers one-click “revalidate and complete.” Final pane shows baseline versus guarded workflow-fidelity scores.

2. Merchant-side stale-state and malicious-content firewall

Problem stated by sources. The July 2026 Decision-Centered Reference Architecture says protocol-valid messages cannot guarantee that price, inventory, policy, and checkout premises remain valid. It explicitly states that cryptographic mandates do not protect against prompt injection or malicious product content, and that production security, live interoperability, performance, legal compliance, and real-world false-positive/negative rates remain untested. ([3.1], [3.2], [3.3])

This is not hypothetical: the 2025 agentic-payments census documents prompt manipulation causing unauthorized financial actions, including a reported **\$47,000 Freysa loss**. AP2 execution-layer research likewise says semantic and intent-level prompt injection remains an open challenge outside replay/context-binding verification. ([4.1], [13.1], [14.1], [14.2])

Track fit. AI Risk Manager: “**stop the merchant losing money to fraud, returns, chargebacks — detection/defense only.**” The product defensively blocks a payment-provider call when catalog content is hostile or the commercial state changed; it neither creates attacks nor verifies shopper identity or spending mandates.

Build/evaluation resources. Fork the public [Agentic Commerce Blueprint](#), whose frozen release includes scenario and ablation fixtures, and use the [OWASP GenAI Security Project](#) taxonomy for defensive test cases. The Blueprint’s suggested metrics include stale-dependency rate, changed-state prevention, blocked-payment-before-provider rate, evidence conflicts, revalidation latency, and remediation time. ([3.3])

One-week MVP. Hash protected checkout dependencies; isolate untrusted product-page text; require a fresh structured merchant decision immediately before creating a Razorpay order/payment; show the exact reason for a block. Use a pretrained LLM only to classify or summarize suspicious instructions, with deterministic rules controlling payment.

30-second demo. A product description silently changes from normal text to an instruction telling the agent to ignore the quoted total. Simultaneously, inventory or price changes. The unguarded flow proceeds; the firewall highlights hostile text and changed hashes, blocks before provider invocation, refreshes state, and permits only the corrected amount.

3. Evidence-grounded multi-document reconciliation copilot

Problem stated by sources. *FinBalance* (June 2026) reports a best exact final-balance-sheet score of only **46%** across six contemporary models. Four models showed a **26–41 percentage-point** gap between reconstructed and self-reported balances; **52%** of otherwise numerically correct journal entries cited the wrong document. Prompting for citations improved this by only 1.4 points, indicating a retrieval/disambiguation problem rather than simple instruction noncompliance. ([5.1], [5.4])

The source explicitly says document-linking failures dominate and that its clean synthetic documents omit production messiness such as OCR errors, handwriting, inconsistent names, and institution-specific artifacts. It also excludes complete GAAP/IFRS, materiality, and audit coverage. ([5.5], [5.2], [5.3])

Track fit. AI Finance Controller, specifically “**reconciliation, exception handling.**” This candidate links each proposed posting to its invoice, bank row, memo, and ledger evidence, refusing closure when documents conflict.

Build/evaluation resources. Use [FinBalance](#) for multi-document, contradiction, linkage, and exact-balance evaluation. Add [FinMaster](#) or its [paper/data description](#) for broader finance workflows; FinMaster reports complex multi-step task accuracy falling to roughly 40%. ([5.1], [5.4])

One-week MVP. Retrieval over document chunks plus deterministic double-entry checks. Require document IDs and quoted evidence for every line item. Replay entries through a small ledger rather than trusting the model’s stated ending balance. No fine-tuning is needed.

30-second demo. Drop four documents into the UI. It draws links from invoice and bank row to a proposed journal entry. One plausible but wrong memo link flashes red; the ledger replay exposes a balance mismatch; selecting the correct source resolves both and closes the period with an evidence packet.

4. Many-to-many cash-application exception resolver

Problem stated by sources. A November 2025 multi-agency ERP reconciliation study documents rule-based matching at no more than 70%, backlogs above 5,000 items, T+3–T+10 cycles, and unresolved batched-remittance, partial-settlement, and FX-rounding exceptions. It reports many-to-many split payments as **28% of errors**, followed by FX drift (22%), dirty references (19%), and duplicate postings (16%). ([6.1], [6.4])

The authors explicitly identify the lack of open, reproducible multi-bank/multi-currency datasets; one payment clearing dozens of invoices remains difficult, and bank-format changes cause **2–5 percentage-point recall degradation** when production feedback is sparse. The proposed anomaly detector also operates at only 10–20% precision for 80–90% recall, leaving a substantial triage problem. ([6.2], [6.3])

A separate July 2025 study says exception handling, scalability, ERP integration, long-term performance measurement, and predictive AI remain inadequately established, despite reducing exceptions from 22% to 8% in one deployment. ([15.1], [15.2])

Track fit. AI Finance Controller: “reconciliation, exception handling.” This is not generic fraud prediction; it resolves cash-application ambiguity and routes only uncertain cases to a controller.

Build/evaluation resources. Generate reproducible payment/invoice streams with the [FinMaster FinSim benchmark](#), use [FinBalance](#) for evidence-linking tests, and ingest test transactions through [Razorpay Payments](#), [Route, Settlements, and Webhooks APIs](#). Because the cited study explicitly says no adequate open multi-agency dataset exists, synthetic benchmark generation is the honest public-data route—not a claim that proprietary bank data is available. ([6.3])

One-week MVP. Candidate generation using amount/date/reference similarity; constrained subset matching for one-to-many and many-to-many cases; LLM explanations over the top matches; confidence-based human queue. Evaluate exact allocation, false auto-post rate, coverage, and analyst clicks.

30-second demo. A ₹1,84,500 remittance with a damaged reference arrives against twelve invoices, two partial credits, and an FX difference. The UI animates candidate allocations, auto-posts the high-confidence group, and places one ambiguous residual in a human queue with a concise explanation and audit trail.

5. Agent-readable catalog and checkout-readiness linter

Problem stated by sources. The 2026 agent-commerce architecture literature distinguishes a product being discoverable from being checkout-ready: syntactically correct protocol data does not establish current commercial eligibility. It calls for live protocol/merchant evaluation and says current fixtures do not establish real-world error rates. ([3.1], [3.2], [3.3])

Towards an Agent-First Web (June 2026) explicitly identifies six open problems, including standards fragmentation, falsifiable capability declarations, small-publisher implementation barriers, and missing cross-agent provenance and billing standards. It warns that infrastructure cost can create a two-tier web excluding small publishers and recommends platform-level tooling. ([8.1])

Track fit. AI Growth & Agentic Commerce: “grow merchant revenue, make merchants sellable to AI buyers.” This directly helps a small merchant expose products, variants, inventory, returns, delivery promises, and checkout actions in machine-readable form. It does not duplicate agent identity or spend controls.

Build/evaluation resources. Use the public [Amazon Shopping Queries/ESCI dataset](#) to test whether agents retrieve the correct product for attribute-rich queries; validate markup against [Schema.org Product](#) and [Google’s Rich Results Test](#). Generate an adapter compatible with the public [Agentic Commerce Blueprint](#). ([3.3], [8.1])

One-week MVP. URL crawler → product/offer/variant extraction with an existing LLM → deterministic completeness checks → generated JSON-LD plus an agent-facing catalog endpoint and simulated checkout action. Score retrieval before and after remediation on an ESCI subset.

30-second demo. Paste a messy merchant product URL. The scorecard shows missing variant IDs, stale availability, ambiguous return policy, and no machine checkout action. Click “repair”; the agent now finds the right SKU for a natural-language query and successfully reaches a Razorpay test checkout.

6. Pre-chargeback evidence and clarification workspace

Problem stated by sources. A 2026 pre-chargeback-network patent explicitly criticizes the current dispute process as consuming bandwidth and network resources, slowing processing, increasing traffic, and contributing to issuer declines. Its proposed direction is direct merchant–cardholder questions and information exchange before formal processor escalation. This is a valid gap statement because the patent expressly describes shortcomings in the existing process. ([9.1])

Agentic-commerce research adds that dispute resolution lacks audit mechanisms for accountability and refunds and that systems still need linked receipts, dispute routes, and portable evidence. ([1.1], [1.3], [10.1])

Track fit. Primary: **AI Revenue Recovery**, because it can preserve revenue by resolving service or evidence problems before a formal chargeback. Secondary: **AI Risk Manager**, under “**stop the merchant losing money to ... chargebacks — detection/defense only.**” This is evidence organization and communication—not generic chargeback-risk prediction or automated representment scoring.

Build/evaluation resources. Use the public [CFPB Consumer Complaint Database and API](#) for complaint narratives and resolution-status fields. Use [Razorpay Disputes API](#) structures to model evidence packets and a mock pre-dispute stage. Evaluate issue classification, required-evidence recall, unsupported-claim rate, time to complete a packet, and simulated resolution rate.

One-week MVP. Complaint classifier, missing-evidence checklist, timeline builder from order/payment/refund events, and grounded draft messages. Keep all external sending human-approved.

30-second demo. A customer says “refund never received.” The app reconstructs payment, delivery, cancellation, and refund events; discovers that the refund was initiated but the customer lacks an ARN/status explanation; drafts a grounded clarification, requests one missing document, and changes the case from “formal dispute likely” to “resolved pre-chargeback.”

7. POS terminal attestation anomaly console

Problem stated by sources. Block’s 2025 payment-terminal patent says terminals remain prime fraud targets; passive eavesdropping and active signal injection can occur without physical access, traditional tamper switches/meshes can be bypassed, applications may lack isolation, and terminals lack visibility into simultaneous payment activity. It expressly acknowledges that complete tamper-proofing is impossible, motivating continuous tamper detection and logical attestation. ([11.1], [11.2], [11.3])

The system also identifies a practical unresolved operating need: dynamically update attestation criteria from improperly accepted/denied transaction feedback rather than rely on static validation. ([11.4], [11.5])

Track fit. **AI Risk Manager:** “**stop the merchant losing money to fraud ... detection/defense only.**” The product monitors merchant payment endpoints and quarantines suspicious devices; it does not generate terminal attacks.

Build/evaluation resources. Use public network/IoT anomaly datasets [IoT-23](#) and [TON IoT](#), plus synthetic terminal-attestation events based on the patent’s event model. Use pretrained anomaly detectors or Isolation Forest; do not train a deep model. Evaluate time-to-detect, false alarms, and percentage of simulated payments blocked after integrity failure.

One-week MVP. Stream device measurements, app/version hashes, location changes, reader pairings, and network anomalies into a terminal trust score. A deterministic policy quarantines a device when cryptographic or high-severity conditions fail; ML only prioritizes softer anomalies.

30-second demo. A live terminal fleet is green. One terminal changes app hash and exhibits anomalous network behavior. Its trust graph turns amber then red, the next test payment is held, and the console displays the exact failed attestation and recommended remediation.

Recommendation

Build candidate 1 if the goal is the strongest combination of novelty, direct agentic-commerce relevance, public benchmark availability, and visual drama. It addresses a measured problem—agents that appear successful while corrupting transaction state—and naturally demonstrates Razorpay payment/refund integration.

Build candidate 2 if competing specifically in AI Risk Manager. It is sharply differentiated from generic fraud scoring because it protects the merchant's own checkout boundary against stale decisions and untrusted content.

Build candidate 3 if implementation certainty matters most. It has the cleanest benchmark, objective metrics, and a highly credible one-week architecture, although finance-document copilots are a more crowded presentation category. Candidate 4 is commercially strong but needs disciplined synthetic-data construction because the literature itself says an adequate open multi-bank dataset is missing. ([5.1], [6.3])

The weakest fit for a one-week solo submission is candidate 7: the unresolved problem is well documented, but a convincing terminal simulation requires more infrastructure and has less immediate relevance to an online Razorpay checkout than the top four.

References

[1.1] From Super-Apps to Agent Economies: Delegated AI Requires Transaction Closure. Chaoyue He, Xin Zhou, Di Wang, Hong Xu, Wei Liu, Chunyan Miao. MDPI AG (2026). <https://doi.org/10.20944/preprints202604.1860.v1> Context: "The paper identifies critical gaps in transaction closure for AI-native agents and delegated systems. While systems like Codex demonstrate strong action closure through orchestration and tool use, they lack surrounding transaction frameworks including payment capture, contract authority, counterparty identity verification, and liability semantics. The document emphasizes that "action closure can mature before payment, contract, liability, and dispute frameworks exist." Key missing elements across AI-native agents and ecosystems include: reversible commitment lifecycles, cancellation and refund call-backs, dispute routing with linked receipts, agent-readable receipts, audit export capabilities, portability safeguards, cross-vendor settlement mechanisms, and agent-specific authority objects. The paper treats super-apps as a stress test for broader transaction-closure problems and argues that closure requirements—rather than historical origin—should determine readiness for agent economies. Table 4 explicitly documents these gaps across different system types."

[1.2] From Super-Apps to Agent Economies: Delegated AI Requires Transaction Closure. Chaoyue He, Xin Zhou, Di Wang, Hong Xu, Wei Liu, Chunyan Miao. MDPI AG (2026). <https://doi.org/10.20944/preprints202604.1860.v1> Context: "This excerpt identifies a critical research gap: current agent evaluation focuses on task completion but neglects transaction closure dimensions including authorization bounds, split-state detection, commitment reversal, receipt preservation, and human recovery burdens. The authors propose ClosureBench, a benchmark comparing identical workflows across super-apps, AI-native surfaces, enterprise suites, open-web stacks, and protocol-native markets under uniform mandates and failure injections (price drift, booking-payment splits, revocations, prompt injections, identity mismatches). Key unresolved gaps include: distinguishing optimal-path from recovery-capable agents; evaluating environment-derived risk isolation; testing resilience against realistic exceptions; and assessing receipt completeness's impact on principal trust. The framework yields five testable predictions about environment quality, scoped delegation, receipt structures, and ecosystem maturity. Missing infrastructure includes: typed mandates, reversible commitment lifecycles, receipt graphs, auditable handoff semantics, and liability-aware recovery mechanisms. Publication date: April 2026."

[1.3] From Super-Apps to Agent Economies: Delegated AI Requires Transaction Closure. Chaoyue He, Xin Zhou, Di Wang, Hong Xu, Wei Liu, Chunyan Miao. MDPI AG (2026). <https://doi.org/10.20944/preprints202604.1860.v1> Context: "The paper identifies critical unresolved gaps in agent economy infrastructure across multiple dimensions. Key missing capabilities include: mandate objects, settlement callbacks, receipt graphs, dispute routes, and liability semantics for tool interoperability; counterparty verification, identity providers, payment rails, revocation propagation, and cancellation/refund semantics for agent transactions; and exportable evidence for authority, action, and recovery. Current evaluation methods are inadequate—focusing on task completion rather than mandate adherence, unauthorized commitment avoidance, split-state detection, commitment reversal, receipt preservation, and human recovery burden minimization. The authors propose ClosureBench, a benchmark design for comparative substrate evaluation that would run identical workflows across super-apps, AI-native surfaces, enterprise suites, open-web stacks, and protocol-native markets to assess transaction-closure capabilities. The paper (April 2026) emphasizes that payment infrastructure alone is insufficient without embedded lifecycle management including authority verification, fulfillment state, reversal, dispute handling, liability frameworks, and portable receipts."

[1.4] From Super-Apps to Agent Economies: Delegated AI Requires Transaction Closure. Chaoyue He, Xin Zhou, Di Wang, Hong Xu, Wei Liu, Chunyan Miao. MDPI AG (2026). <https://doi.org/10.20944/preprints202604.1860.v1> Context: "The excerpt identifies critical gaps in agentic commerce infrastructure, particularly the need to move from task-completion benchmarks toward portable transaction-closure systems. Key unresolved problems include: implementing auditable receipt graphs, commitment reversibility, and tamper-evident handoff packets; addressing security threats from prompt-injection attacks, mali-

cious merchants, and over-delegation; establishing least-privilege mandate enforcement and effective revocation propagation; and creating principal-facing control interfaces that balance expressiveness with usability. The authors note that typed mandates, commitment lifecycles, and dispute resolution mechanisms must be extracted from existing operational environments and embedded into open-standard layers. They also flag the tension between comprehensive closure and privacy/contestability concerns. The publication is dated April 2026."

[2.1] Beyond Task Success: Measuring Workflow Fidelity in LLM-Based Agentic Payment Systems. Donghao Huang, Joon Kiat Chua, Zhaoxia Wang. Lecture Notes in Computer Science (2026). https://doi.org/10.1007/978-981-92-2014-4_29 Context: "The excerpt addresses workflow fidelity gaps in LLM-based payment agents. It identifies a security vulnerability where 10 of 18 tested models skip mandatory CPA-mediated confirmation checkpoints during payment processing, creating audit trail gaps that violate PCI-DSS compliance requirements. These deviations occur despite achieving near-perfect task success metrics, remaining invisible to conventional evaluation methods. The gap is systematic rather than random—all affected models follow an identical 9-step shortcut pattern triggered by direct payment phrases, indicating interaction between input phrasing and model reasoning. The paper introduces ASR (Action Sequence Recall) as a trajectory-level metric to detect such hidden workflow deviations. However, the excerpt does not discuss indirect prompt injection, malicious merchant content, unintended consequences, or transaction recovery mechanisms. No public benchmarks, datasets, or code repositories are named in this excerpt."

[3.1] A Decision-Centered Reference Architecture for Trustworthy Agentic Commerce. Dimitrios S. Sfiris. ArXiv (2607). <https://doi.org/10.48550/arxiv.2607.18347> Context: "This excerpt presents a Decision-Centered Reference Architecture for agentic commerce that addresses the merchant-side decision layer. Key unresolved gaps explicitly stated include: rule completeness, production security, performance, legal compliance, live interoperability, population error rates, and independent replication. The work does not claim to test rule completeness or production-grade security. Quantitative results show that all seven initially valid actions were blocked after protected state changed, with 66/66 tests passing under synthetic conditions. The paper emphasizes that protocols standardize external exchange but merchants need an authoritative internal representation of commercial eligibility, actor authority, and checkout validity. The WebDigestPro guide (updated 15 July 2026) publicly describes the decision-centered architecture elements. The evaluation was limited to synthetic ecommerce scenarios with controlled ablations, not live production interoperability or real-world error rates. The work explicitly identifies that performance benchmarking, legal compliance verification, and population-scale testing remain untested."

[3.2] A Decision-Centered Reference Architecture for Trustworthy Agentic Commerce. Dimitrios S. Sfiris. ArXiv (2607). <https://doi.org/10.48550/arxiv.2607.18347> Context: "This excerpt identifies several unresolved gaps in agentic commerce systems. The primary architectural gap is semantic drift across protocol boundaries—while protocols like ACP, AP2, UCP, and MCP are syntactically valid, they cannot guarantee commercial premise validity, freshness, or authorization without merchant-side decisions. Key limitations identified include: cryptographic mandates alone cannot protect against prompt injection or malicious product content; replay and context-binding failures exist at execution time; and independently interpreting catalog flags and policy pages creates conflicting decisions across layers. The paper emphasizes that reliability depends on combined behavior of multiple systems (AI, merchant data, identity, checkout, payment, protocols) rather than model accuracy alone. The reference implementation is explicitly stated as not production-deployed. No quantitative results, public code links, or dataset references are provided in these pages."

[3.3] A Decision-Centered Reference Architecture for Trustworthy Agentic Commerce. Dimitrios S. Sfiris. ArXiv (2607). <https://doi.org/10.48550/arxiv.2607.18347> Context: "# Summary This excerpt describes the "Agentic Commerce Blueprint," a protocol-agnostic decision architecture for trustworthy AI agent transactions in e-commerce. The work addresses **whether merchants can reliably decide if an AI agent's requested action is commercially valid, authorized, current in checkout state, and supported by verified evidence** before payment proceeds. **Key unresolved gaps explicitly stated:** The research agenda (Section 7.5) lists critical gaps: "Independent replication," "external validity and false rejection" evaluation with real merchant data, "live ACP, AP2, UCP, or MCP adapters" integration, "production experiments measuring latency, throughput, concurrency

ances," and "legal and human-factors evaluation" of mandate comprehension and dispute reconstruction. The limitations (7.4) acknowledge the study used "synthetic, not anonymized merchant transactions" and "does not estimate prevalence, false-positive or false-negative rates, customer impact, exploitability, or independent reproducibility." **Demonstrated capability:** The reference implementation (v0.9.2, publicly available on GitHub) passed 66/66 tests, prevented stale-decision execution, and enforced verified-state identity through detached cryptographic envelopes and protected-dependency revalidation—but only within controlled ablations, not production or live payment-provider integrations."

[4.1] The rise of Agentic Payments: a protocol census and analysis of early market solutions. E Donadello, L ZEREDO PEREIRA. 2025.

https://www.politesi.polimi.it/bitstream/10589/260157/1/2026_07_Donadello_Zeredo%20Pereira.pdf Context: "The excerpt identifies several unresolved challenges and gaps in agentic payments and commerce. Regarding liability and authorization, the text explicitly states that agentic payments introduce 'new questions about identity, authorisation, liability, and consumer protection' that require exploration. The excerpt highlights authorization ambiguity and unresolved liability as essential legal challenges in agentic payments, though this section appears to be cut off mid-sentence. The document also discusses the black box problem in AI agents, noting that LLM-based agents can produce 'hallucinations' and are vulnerable to prompt manipulation, citing the Freysa AI agent incident where external prompts manipulated the agent into executing unauthorized transactions. Additionally, the excerpt identifies data quality issues, regulatory compliance gaps under the EU AI Act, and systemic risk concerns. However, the excerpt does not explicitly mention transaction lifecycle closure, refunds/disputes, receipts, workflow fidelity, merchant interfaces, evaluation benchmarks, datasets, or code repositories. The publication date is 2025."

[5.1] FinBalance: A Multi-Document Accounting Reconciliation Benchmark. Sasank Tumpati, D. Agarwal, Ayush Kedia, Arjun Neekhra, Murari Mandal, Krishna Garg, Yash Sinha, Suman Gupta, Dhruv Kumar. ArXiv (2026). <https://doi.org/10.48550/arxiv.2606.15949> Context: "The FinBalance benchmark reveals several explicit limitations and measured gaps in LLM performance on accounting reconciliation tasks. The highest exact final-balance-sheet score achieved across six evaluated models (Gemini 3 Flash, GPT-5, Claude Haiku 4.5, Grok-4.3, DeepSeek Chat, Qwen 3 235B) is only 46%, with strict end-to-end accuracy remaining low. A self-consistency aggregation gap exists where four of six models show 26-41 percentage point differences between reconstructed balance sheets and self-reported ones. The dominant failure mode is document-linking: 52% of journal entries with correct accounts and amounts cite wrong documents, and this barely improves (-1.4 pp) with explicit citation-pressure prompting, indicating retrieval-disambiguation challenges rather than prompt compliance issues. Diagnostic tool feedback demonstrates trade-offs: providing balance-sheet deltas improves BSexact by 30-33 pp but degrades inconsistency-code detection by 13-52 pp on smaller models. The evaluation covers 710 records and includes failure breakdowns by industry, ledger family, concept flags, and error types."

[5.2] FinBalance: A Multi-Document Accounting Reconciliation Benchmark. Sasank Tumpati, D. Agarwal, Ayush Kedia, Arjun Neekhra, Murari Mandal, Krishna Garg, Yash Sinha, Suman Gupta, Dhruv Kumar. ArXiv (2026). <https://doi.org/10.48550/arxiv.2606.15949> Context: "The excerpt describes explicit limitations of FinBalance (a multi-document accounting reconciliation benchmark). Key unresolved gaps include: (1) Document fidelity - synthetic rendered documents don't approximate real-world messiness (OCR errors, handwriting, inconsistent naming, institution-specific artifacts); (2) Scope limitations - the benchmark covers multi-document postings, period-end adjustments, tax, FX, leases, deferred tax, revenue schedules, and subledgers but excludes full GAAP/IFRS coverage, materiality judgments, and audit procedures; (3) Error analysis reveals document-linking failures dominate (93.1% of 480 standard records), indicating grounding plausible entries to correct evidence remains the hard part. The benchmark measures reasoning over clean OCR text rather than end-to-end document understanding in production systems. No information about FinMaster, Finch, or other payment reconciliation benchmarks appears in this excerpt."

[5.3] FinBalance: A Multi-Document Accounting Reconciliation Benchmark. Sasank Tumpati, D. Agarwal, Ayush Kedia, Arjun Neekhra, Murari Mandal, Krishna Garg, Yash Sinha, Suman Gupta, Dhruv Kumar. ArXiv (2026). <https://doi.org/10.48550/arxiv.2606.15949> Context: "The excerpt discusses FinBalance, a multi-docu-

ment accounting reconciliation benchmark. Key limitations include: the benchmark does not cover the full space of GAAP, IFRS, local tax regimes, materiality judgments, industry-specific policy choices, or audit procedures. Analysis is conducted at behavioral level but does not establish internal mechanisms causing failure modes or whether failures transfer to other high-precision domains (legal reconciliation, insurance claims, scientific records). Identified failure modes include: aggregation gap, weak gains from prompt-only changes, document-linking failures, context/evidence effects, and ledger-feedback trade-off with inconsistency detection. The core evaluation uses six models tested on 710-record core split and 143-record compact split. However, the excerpt does not contain information about FinMaster, Finch, or other payment reconciliation work, nor does it provide explicit code/data links, task counts, or specific model performance metrics."

[5.4] FinBalance: A Multi-Document Accounting Reconciliation Benchmark. Sasank Tumpati, D. Agarwal, Ayush Kedia, Arjun Neekhra, Murari Mandal, Krishna Garg, Yash Sinha, Suman Gupta, Dhruv Kumar. ArXiv (2026). <https://doi.org/10.48550/arxiv.2606.15949> Context: "The excerpt describes FinBalance, a multi-document accounting reconciliation benchmark released in June 2026. Key limitations and gaps identified include: (1) A significant 26-41 percentage point gap between models' reported balance sheets (BSexact) and the correct balance sheet obtained by replaying entries through the ledger (BSrecon), indicating aggregation inconsistencies; (2) Models struggle to bind extracted journal entries to supporting source documents; (3) Document-linking errors persist despite citation-pressure prompting interventions; (4) Only 46% exact final balance-sheet accuracy achieved across six contemporary LLMs on the 710-record evaluation split. The benchmark includes 23 inconsistency-code labels and covers eight industries with three period types and five difficulty levels. Ledger-feedback ablations showed improvements and exposed inconsistency-detection trade-offs. The excerpt does not mention Finch or payment reconciliation work specifically, and does not provide code/data links beyond the ArXiv URL."

[5.5] FinBalance: A Multi-Document Accounting Reconciliation Benchmark. Sasank Tumpati, D. Agarwal, Ayush Kedia, Arjun Neekhra, Murari Mandal, Krishna Garg, Yash Sinha, Suman Gupta, Dhruv Kumar. ArXiv (2026). <https://doi.org/10.48550/arxiv.2606.15949> Context: "The FINBALANCE benchmark (2026) identifies several explicit limitations and gaps: (1) An aggregation gap measured by the difference between BSrecon (entries applied to opening balance) and BSexact (self-reported balance sheet match), indicating discrepancies in how models construct versus report statements. (2) Forced-inconsistency detection challenges, with 32% of the evaluation split dedicated to records where documents are intentionally contradictory—models must recognize inconsistency rather than fabricate reconciled statements. (3) Context-length vulnerability where adding distractors or moving relevant evidence later degrades performance, showing the task remains difficult even when context volume is controlled. (4) Tool effectiveness limitations where passive tools (calculators, search) are underutilized, and while ledger-feedback tools can repair reported balance sheets, they simultaneously weaken the model's ability to detect inconsistencies. The excerpt focuses on FINBALANCE methodology but does not provide explicit performance metrics, availability links, or comparative analysis of FinMaster, Finch, or other reconciliation benchmarks."

[6.1] FINTECH TRANSFORMATION: AI AND RPA BOTS FOR MULTI-AGENCY PAYMENT RECONCILIATION IN ERP. Sravanthi Gondi, Harsh Parnerkar b Kishore Subramanya, Hebbar. International Journal of Applied Mathematics (2025). <https://doi.org/10.12732/ijam.v38i10s.944> Context: "The research identifies critical unresolved problems in multi-agency payment reconciliation across complex ERP systems (SAP S/4HANA, Oracle EBS, Microsoft Dynamics 365, PeopleSoft). Key challenges include: heterogeneous transaction formats (CAMT.053, MT940, BAI2, EDI 820, PDFs) across 325 legal entities and 210 bank partners; rule-based matching achieving only $\leq 70\%$ performance with backlogs $> 5,000$ items; SLA violations $> 12\%$; reconciliation cycle times of T+3 to T+10 days; and write-offs of 0.1-0.3% of accounts receivable. The paper identifies unresolved limitations in exception handling for batched remittances, partial settlements, and FX round-off issues. Governance limitations include audit trail requirements (7-year retention), segregation of duties restrictions, and dual-approval requirements for high-value items. Published November 2025 in the International

Journal of Applied Mathematics. The research proposes AI-RPA solutions achieving 88-95% auto-match rates and 60-95% cycle time reduction, with evaluation metrics including precision >97%, recall >92%, and statistical testing at $\alpha=0.05$."

[6.2] FINTECH TRANSFORMATION: AI AND RPA BOTS FOR MULTI-AGENCY PAYMENT RECONCILIATION IN ERP. Sravanthi Gondi, Harsh Parnerkar b Kishore Subramanya, Hebbar. International Journal of Applied Mathematics (2025). <https://doi.org/10.12732/ijam.v38i10s.944> Context: "The paper documents exception handling mechanisms for RPA-based payment reconciliation in multi-agency ERP systems. Key unresolved challenges include managing exceptions bucketed by confidence, amount, currency, and counterparty risk, with human-in-the-loop queues targeting ≤ 1 day work-in-progress at p90. Anomaly detection uses Isolation Forest and autoencoders, expecting 1-3% anomaly base rates with detectors set at 80-90% recall and 10-20% precision. Exception types addressed include duplications, split payments, FX rounding, weekend cutoffs, and partial settlements. The system requires audit trails with 7-year data retention including inputs, outputs, model versions, and approver signatures. Limitations include managing false positives (<2% target), drift detection via population stability index ($PSI < 0.2$), and automated rollback triggers when accuracy drops >1pp. The evaluation protocol targets 88-95% auto-match rates. Publication date: November 2025."

[6.3] FINTECH TRANSFORMATION: AI AND RPA BOTS FOR MULTI-AGENCY PAYMENT RECONCILIATION IN ERP. Sravanthi Gondi, Harsh Parnerkar b Kishore Subramanya, Hebbar. International Journal of Applied Mathematics (2025). <https://doi.org/10.12732/ijam.v38i10s.944> Context: "The excerpt identifies several unresolved problems in RPA reconciliation systems. Key limitations include: lack of open reproducible datasets for multi-agency reconciliation across different banks, currencies, and document noise; hyperparameter settings that can encode workflow biases; many-to-many matching challenges where single payments clear dozens of invoices creating false positives; sparse user feedback loops in production causing 2-5 percentage point recall degradation when bank formatting changes; need for continuous active learning and recalibration. Exception handling uses fallback rules that forward high-value items, currency misalignments, and duplicates to manual review. Anomaly detection employs Isolation Forest and autoencoders with intentionally low precision (10-20%) at 80-90% recall to maximize coverage. The study involved 8 agencies with 5 banking partners and 7 currencies over 12 months producing ~24 million line items (publication date: November 2025)."

[6.4] FINTECH TRANSFORMATION: AI AND RPA BOTS FOR MULTI-AGENCY PAYMENT RECONCILIATION IN ERP. Sravanthi Gondi, Harsh Parnerkar b Kishore Subramanya, Hebbar. International Journal of Applied Mathematics (2025). <https://doi.org/10.12732/ijam.v38i10s.944> Context: "The article identifies several unresolved problems in RPA reconciliation: many-to-many split payments (28% of errors) remain especially troublesome due to partial payment matching across entities and currencies; currency FX drift (22% of errors) occurs from minor exchange rate fluctuations in cross-border transactions; dirty references (19% of errors) stem from irregular data input quality; and duplicate postings (16% of errors) result from system reprocessing or downtime. The authors acknowledge that exception handling processes require further enhancement to minimize these issues. Audit trails are supported through PeopleSoft's data lineage functionality and real-time compliance anomaly detection. The research was conducted August 2025 using an 8-week A/B testing period comparing AI-driven reconciliation (treatment) against manual rules-based systems (control). The system achieved 97.8% accuracy in offline models and demonstrated 72% cycle time reduction, though human escalation mechanisms for exception resolution are not explicitly detailed beyond mention of remediation playbooks being created."

[7.1] The Relational Accounting Execution Layer: Governed Automation Across Subledger and Cost-Accounting Processes. JY Hunag. 2026. <https://utoronto.scholaris.ca/bitstreams/e94eb640-78ab-483f-964a-68992c38f558/download> Context: "The excerpt identifies several explicit limitations in the RAEL (Relational Accounting Execution Layer) architecture for accounts receivable and cost accounting automation. The architecture is conceptual and lacks empirical evaluation regarding comparative accuracy, processing efficiency, or organizational effectiveness. Automation eligibility is context-dependent, with the proportion of events eligible for automated processing varying across industries and organizations. Implementation requires establishing thresholds for confidence, materiality, tolerance, and escalation within specific organizational contexts, but the archi-

texture does not prescribe universal numerical values. Platform-specific limitations affect complete implementation, including API availability, data structures, access controls, and verification capabilities. The work addresses human escalation through management-by-exception and verified closure mechanisms, with professional judgment embedded in exception pathways. However, quantitative performance benchmarks, implementation guidance across different platforms, and comparative effectiveness data remain unresolved."

[8.1] Towards an Agent-First Web: Redesigning the Web for AI Agents. Eranga Bandara, Ross Gore, R. Mukkamala, A. Gunaratna, S. Bouk, Xueping Liang, Peter B. Foytik, Abdul Rahman, Sachini Rajapakse, Isurunima Kularathna, Pramoda Karunarathna, C. Rajapakse, Ng Wee Keong, K. D. Zoysa, T. Hewa, Amin Hass, W. Herath, A. Withanage, Nilaan Loganathan, Atmaram Yarlagadda, Sachin Shetty. ArXiv (2026). <https://doi.org/10.48550/arxiv.2606.19116> Context: "# Summary This excerpt discusses open challenges in agent-mediated web interaction, not payment systems or fraud detection. The key unresolved problems identified are: 1. **Agent advertising revenue models**: "Revenue sharing requires bilateral agreements between every publisher and every agent frontend operator — a combinatorially complex negotiation landscape" and regulatory mandates face jurisdictional fragmentation. 2. **Copyright ambiguity**: "The copyright status of the synthesized output is legally unsettled in most jurisdictions" when AI agents synthesize copyrighted content. 3. **Data protection liability**: "It is unclear which entity — the user, the agent provider, the content publisher — bears data controller responsibility" under GDPR. 4. **Platform liability classification**: "It is unclear whether AI agents that synthesize and present content to users are acting as platforms...or as publishers." 5. **Small publisher equity**: Implementation costs create infrastructure barriers, potentially creating "a two-tier web in which large publishers dominate agent-accessible content." 6. **Multi-agent coordination**: "A cross-agent provenance standard...does not currently exist" for tracking content across multiple agents. These are payment/commerce infrastructure gaps, not fraud or checkout-specific problems relevant to the Razorpay tracks."

[9.1] Communication network and method for routing data messages on networks having John Larko. Patent (WO,EP,US,MX,RU) (2026). <https://patents.google.com/patent/US12561695B2/en> Context: "The patent describes a pre-chargeback network system designed to resolve transaction disputes between cardholders and merchants before they escalate to formal chargebacks through payment processors. The technical problem being addressed is inefficiency in the dispute resolution process: the current chargeback system consumes network resources and bandwidth, slows payment processing, generates excessive network traffic, and leads to issuer declines. The system enables direct communication between disputing parties to submit questions and request information, allowing disputes to be resolved more easily without involving the payment processor. This approach aims to improve payment network efficiency, reduce processing delays, enhance data accuracy, and decrease the volume of chargebacks entered into the system. The patent is from 2026 (US12561695B2) and focuses specifically on transaction dispute resolution as an alternative to traditional chargeback processes."

[10.1] TessPay: Verify-then-Pay Infrastructure for Trusted Agentic Commerce. Mehul Goenka, Tejas Pathak, Siddhartha Asthana. ArXiv (2602). <https://doi.org/10.48550/arxiv.2602.00213> Context: "The TessPay paper identifies critical gaps in agentic commerce infrastructure. Key unresolved limitations include: (1) task delegation lacks mechanisms to define scope and authorize actions based on user intent; (2) verified settlement uses optimistic models without proof of execution; (3) dispute resolution lacks audit mechanisms for accountability and refunds; (4) existing protocols (Google AP2, Coinbase x402, ACP, ERC-8004) fragment the landscape with no unified solution addressing end-to-end trust. Current approaches rely on optimistic 'pay-then-consume' models or reputation scoring, while tight coupling of reasoning with key custody exposes users to prompt injection risks and unauthorized fund access. The paper proposes TessPay's dual-plane architecture (Control/Verification and Payment Settlement) to close these gaps through cryptographic evidence capture and conditional verified settlement."

[11.1] Logical validation of devices against fraud and tampering. Dino Dai Zovi. Patent (US) (2025). <https://patents.google.com/patent/US12367501B2/en> Context: "This patent (US12367501B2, 2025) explicitly addresses payment terminal security and attestation. The problem statement identifies that running multiple applications simultaneously on computer platforms creates isolation and protection vulnerabilities. The patent notes that modern operating system source code is too large to verify correctness, making it difficult to ensure

expected behavior. A critical problem is ensuring that POS terminals, payment applications, and card readers remain un-tampered, un-modified, and secure throughout payment processing. The patent acknowledges that complete tamper-proofing is impossible, necessitating tamper-resistant or tamper-detecting approaches instead. The solution involves attestation routines using trust commands, integrity metrics, and attestation tickets to establish trusted operation of payment platforms before processing sensitive financial data. This directly addresses payment terminal attestation capabilities."

[11.2] Logical validation of devices against fraud and tampering. Dino Dai Zovi. Patent (US) (2025). <https://patents.google.com/patent/US12367501B2/en> Context: "Patent US12367501B2 (filed 2016, issued 2025) addresses payment terminal security and fraud detection. The explicit problem statements include: (1) payment terminals are prime targets for attackers attempting to access payment information and process fraudulent transactions; (2) attackers can eavesdrop on signals (passive attacks) or inject malicious signals (active attacks) without physical access; (3) traditional tamper detection measures like tamper switches and meshes can be bypassed; (4) payment terminals lack visibility into other payment transactions occurring simultaneously, limiting fraud detection capabilities; (5) applications running on payment platforms are not isolated from one another, creating security vulnerabilities. The patent indicates the industry is solving the capability of attesting payment devices and validating sessions between devices to determine security for financial information exchange. The problem explicitly acknowledges limitations in existing physical and logical fraud/tampering detection methods."

[11.3] Logical validation of devices against fraud and tampering. Dino Dai Zovi. Patent (US) (2025). <https://patents.google.com/patent/US12367501B2/en> Context: "US patent US12367501B2 (filed 2024, published 2025) by Dino Dai Zovi and inventors at Block Inc addresses payment terminal security and fraud prevention. The patent explicitly identifies multiple problems: payment terminals face attacks from third parties attempting to access payment information and process fraudulent transactions through physical component access or signal eavesdropping. Attackers can bypass traditional tamper switches and tamper meshes, create counterfeit EMV cards, and construct devices that intercept or modify signals between terminals and payment devices. Additionally, payment terminals lack visibility into other concurrent payment transactions, limiting fraud detection capabilities. The patent indicates the industry is solving the capability need to attest that payment entities (POS terminals, payment applications, card readers) remain untampered, unmodified, and secure throughout transaction processing. The solution employs augmented tamper and fraud detection methodology through trust routines, device attestation, secure session validation, and attestation tickets to establish device trustworthiness for predetermined periods."

[11.4] Logical validation of devices against fraud and tampering. Dino Dai Zovi. Patent (US) (2025). <https://patents.google.com/patent/US12367501B2/en> Context: "US patent 12367501B2 (2025) by Dino Dai Zovi addresses logical validation of payment terminals against fraud and tampering. The patent describes a system where a payment server generates attestation routines with test criteria to validate payment terminals. The system detects fraudulent transactions and tamper attempts by analyzing attestation data alongside payment transaction data. Notably, the patent includes a feedback loop mechanism where merchants can indicate improperly denied or accepted transactions, which then triggers updates to the test criteria through machine learning techniques. The system is triggered by various conditions including new reader pairing, payment application installation, device relocation, fraudulent card detection, and geo-fence violations. The patent indicates the industry is solving the problem of dynamically validating payment terminal integrity and improving fraud detection accuracy through iterative feedback mechanisms rather than static validation rules."

[11.5] Logical validation of devices against fraud and tampering. Dino Dai Zovi. Patent (US) (2025). <https://patents.google.com/patent/US12367501B2/en> Context: "The patent (US12367501B2, 2025) addresses device attestation and tamper detection for payment systems, particularly POS terminals. The patent indicates the industry is solving the problem of verifying whether hardware or software has been modified from factory configuration, and detecting unauthorized changes including users tampering with software to circumvent technological protections. The document describes implementing software anti-tamper techniques including tamper resistance, detection, response, and evidence collection. It also discusses impeding attack discovery and initial exploitation through obfuscation and adaptation/dynamism to counter deployed attacks. The patent applies these

concepts to brick-and-mortar retail POS terminals and self-checkout systems, with extensibility to e-commerce locations. The technology separates attestation processes from secure session establishment and uses service level agreements (SLAs) to prioritize high-priority processes like generating attestation tickets over lower-priority tamper detection processes."

[12.1] One Success Isn't Reliability: Thinkingbox, a Sandbox and Benchmark for Agents in Stateful Business Workflows. Zhuochun Li, Youngmin Ko, Ali Keramati, Nicola Ferri, Susana Palmaz Lopez Pelaez, Liang-Chun Tsai, Calvin Wang, Mirco Milletari, Tuhin Kundu, V. Smolyakov, Kjartan Ólafsson, Tommy Guy. ArXiv (2026). <https://doi.org/10.48550/arxiv.2608.19741> Context: "# Summary The excerpt introduces **THINKINGBOX**, a sandbox and benchmark for evaluating AI agents in stateful business workflows. Key findings: **Core Problem:** Agents struggle with multi-turn, consequence-bearing tasks requiring coordination of user interaction, tool use, policy constraints, and backend side effects. A "discovery-reliability gap" exists: the strongest model achieves 65.36% pass@1 but only 25.25% pass^20 (success on all 20 attempts), despite 91.12% passing at least once—meaning "successful trajectories are often discoverable but not reliably repeatable." **THINKINGBOX-BENCH contains 507 policy-conditioned workflows** across five domains: retail/e-commerce, travel/hospitality, auto insurance, neobank internal IT, and consulting IT/HR support. Tasks are evaluated by "executable checks that accept valid trajectories while rejecting wrong, missing, or extra effects," measuring terminal backend state changes, not just dialogue. **Critical Gap Stated:** "Many failed trials show clean termination and valid state-changing actions, showing that response or tool-call-level signals are not clear proxies for end-to-end task completion." The benchmark is publicly released: <https://github.com/microsoft/thinkingbox> (dated August 2026)."

[13.1] The rise of Agentic Payments: a protocol census and analysis of early market solutions. E Donadello, L ZEREDO PEREIRA. 2025. https://www.politesi.polimi.it/bitstream/10589/260157/1/2026_07_Donadello_Zeredo%20Pereira.pdf Context: "The excerpt mentions specific security incidents involving AI agents that demonstrate vulnerabilities to prompt manipulation. The Freysa AI agent experienced a \$47,000 loss due to a security flaw allowing external prompts to manipulate it into executing unauthorized transactions. Additionally, a separate incident involved an LLM agent losing \$2,500 after being manipulated into generating phishing content during a routine task. These examples illustrate risks of prompt injection and malicious manipulation of AI agents in financial contexts. However, the excerpt does not provide explicit unresolved security gaps categorized by type (indirect prompt injection, malicious merchant content, workflow fidelity, etc.), nor does it cite named public benchmarks, datasets, or code repositories with specific dates. The document appears to end mid-section at 'Authorization ambiguity and unresolved liability,' suggesting the complete analysis of agentic commerce challenges may continue beyond page 41."

[14.1] Zero-Trust Runtime Verification for Agentic Payment Protocols: Mitigating Replay and Context-Binding Failures in AP2. Qianlong Lan, Anuj Kaul, Shaun Jones, Stephanie Westrum. ArXiv (2602). <https://doi.org/10.48550/arxiv.2602.06345> Context: "The Zero-Trust Runtime Verification (ZTRV) work for AP2 identifies several key limitations and gaps. It does not address physical or side-channel attacks on cryptographic infrastructure, and cannot prevent intent-level attacks such as prompt injection that manipulate legitimate agents into executing unintended mandates. The authors explicitly note that semantic and intent-level attack mitigation remains an open challenge requiring complementary defenses at agent reasoning and intent-validation layers beyond execution-layer verification. Performance-wise, verification state scales with peak concurrency rather than cumulative transaction history, with overhead dominated by cryptographic operations. The paper proposes future integration of Trusted Execution Environments (TEEs) as a hardening mechanism. No public code or dataset links are provided in this excerpt, and quantitative performance results are referenced but not detailed in these pages."

[14.2] Zero-Trust Runtime Verification for Agentic Payment Protocols: Mitigating Replay and Context-Binding Failures in AP2. Qianlong Lan, Anuj Kaul, Shaun Jones, Stephanie Westrum. ArXiv (2602). <https://doi.org/10.48550/arxiv.2602.06345> Context: "The Zero-Trust Runtime Verification (ZTRV) framework for AP2 agentic payments has several explicit limitations and operational considerations. Key unresolved gaps include: (1) inability to prevent prompt injection attacks or authorized misuse where legitimate agents are ma-

nipulated to execute mandates for unintended purposes; (2) assumes cryptographic primitives remain uncompromised and does not address physical or side-channel attacks; (3) requires reliable clock synchronization (NTP) to prevent valid mandates from being rejected due to clock drift. Quantitative results show the verifier achieves ~3.7ms latency at 10,000 RPS with memory footprint of several megabytes. The framework is designed for integration as an execution-layer enhancement (API gateway, reverse proxy, service mesh sidecar) without requiring AP2 protocol changes, enabling incremental adoption. However, the work explicitly acknowledges that semantic/intent-level attack prevention remains beyond execution-layer verification scope."

[15.1] RPA for Account Reconciliations: Case Study of 85% Time Reduction.. Anjali Kale. The American Journal of Interdisciplinary Innovations and Research (2025). <https://doi.org/10.37547/tajir/volume07issue07-09> Context: "The excerpt identifies several explicit unresolved problems in RPA reconciliation: inadequately established exception handling mechanisms, restricted ERP system integration, absence of long-term performance indicators, and insufficient AI utilization for predictive intelligence. While the case study reduced exception rates from 22% to 8%, audit trail functionality via BlackLine was implemented for SOX/IFRS compliance. Key limitations include measurement within a specific organizational context that may not generalize across different ERP systems, study scope limited to a single department with developing automation maturity, and lack of structured longitudinal surveys for assessing workforce transformation. The research was published in July 2025 in The American Journal of Interdisciplinary Innovations and Research. The authors propose a seven-stage RPA Reconciliation Framework addressing these gaps. Recommended future work includes establishing cross-industry RPA maturity benchmarks, evaluating AI-integrated reconciliation models, and investigating organizational change dynamics during extended RPA lifecycles."

[15.2] RPA for Account Reconciliations: Case Study of 85% Time Reduction.. Anjali Kale. The American Journal of Interdisciplinary Innovations and Research (2025). <https://doi.org/10.37547/tajir/volume07issue07-09> Context: "The excerpt identifies several unresolved problems in RPA implementation for account reconciliation. Key limitations include deficiencies in exception handling, scalability, and ERP system integration. The research reveals gaps in empirical specificity and post-implementation evaluations. The paper reviews literature from 2018-2024 and presents a case study of a multinational life sciences firm using Oracle Cloud Financials, BlackLine automation software, and UiPath bots. Unresolved issues include inadequate quantifiable value assessments, workflow complexity understanding, and deployment pathway clarity. The research recommends future work should focus on hybrid RPA-AI systems and uniform maturity models. Data sources for the case study included project records, performance dashboards, and stakeholder interviews. However, the excerpt does not specifically address exception triage, many-to-many cash application, audit trails, or human escalation mechanisms."